

Intersección σ -álgebra

Proposición 1. Sean Σ_1, Σ_2 dos σ -álgebras en un conjunto $X \neq \emptyset$

$\implies \Sigma_1 \cap \Sigma_2$ es una σ -álgebra en X .

Demostración: Comprobamos las propiedades de la `sigma-algebra`/??:

- (i) Se tiene que $X \in \Sigma = \Sigma_1 \cap \Sigma_2$ porque $X \in \Sigma_1, \Sigma_2$.
- (ii) $E \in \Sigma \implies E \in \Sigma_1, \Sigma_2 \implies E^c \in \Sigma_1, \Sigma_2 \implies E^c \in \Sigma$.
- (iii) Sea $\{E_n\}_{n=1}^\infty \subset \Sigma \implies \{E_n\}_{n=1}^\infty \subset \Sigma_1, \Sigma_2 \implies \bigcup_{n=1}^\infty E_n \in \Sigma_1, \Sigma_2 \implies \bigcup_{n=1}^\infty E_n \in \Sigma$.

Concluimos que $\Sigma_1 \cap \Sigma_2$ es una σ -álgebra en X . ■

Referenciado en

- `prop-sigma-algebra-generada`

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